TEST CODE **02220020**

MAY/JUNE 2012

FORM TP 2012180

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

AGRICULTURE, ENERGY AND ENVIRONMENTAL POLLUTION

UNIT 2 – Paper 02

2 hours 30 minutes

22 MAY 2012 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. Do **NOT** open this examination paper until instructed to do so.
2. This paper consists of **SIX** questions, **TWO** from each Module.
3. Candidates **MUST** answer **ALL** questions.
4. Write your answers in the answer booklet provided.
5. You may use a silent, non-programmable calculator to answer items.

2180

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

NOTHING HAS BEEN OMITTED.

MODULE 1

Answer BOTH questions.

1. (a) State THREE characteristics of subsistence agricultural systems. **[3 marks]**

(b) Discuss, giving THREE reasons, why natural disasters are threats to sustainable agriculture. **[6 marks]**

(c) A student from a Caribbean island assessed the impact of agriculture on the island's environment. The results obtained by the student are shown in Figure 1.

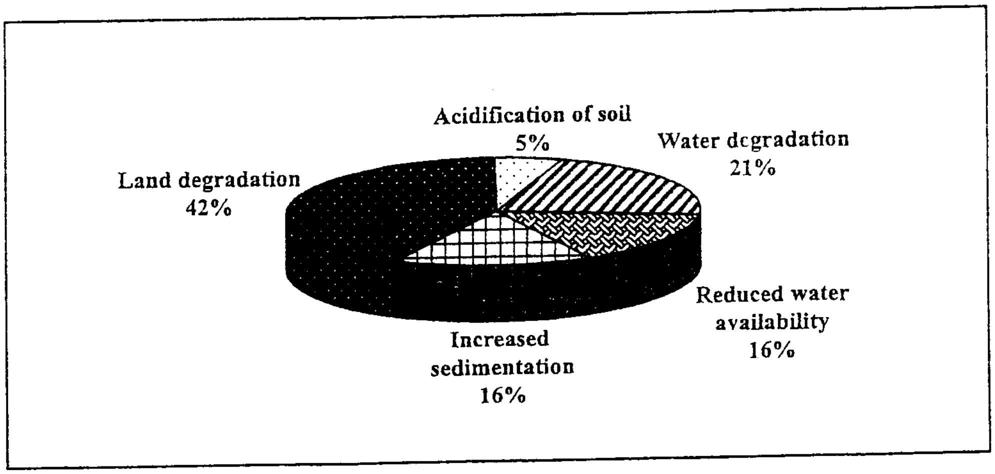

Figure 1. Impact of agriculture on the environment

(i) Study Figure 1 and make THREE deductions about the impact of agriculture on the environment. **[3 marks]**

(ii) Based on the results depicted in Figure 1, what is the total percentage impact on water resources? Show your working. **[2 marks]**

(iii) Explain, giving THREE reasons, why land degradation was as high as 42 per cent. **[6 marks]**

Total 20 marks

2. (a) State FOUR features of sustainable agriculture. **[4 marks]**

(b) Choosing the right pest control method is an integral part of sustainable agriculture. Figure 2 shows the frequency of use of different methods of pest control by three farms, Farm A, Farm B and Farm C. The three methods of pest control used were Integrated Pest Management (IPM), Biological Control (BC) and Chemical Control (CC).

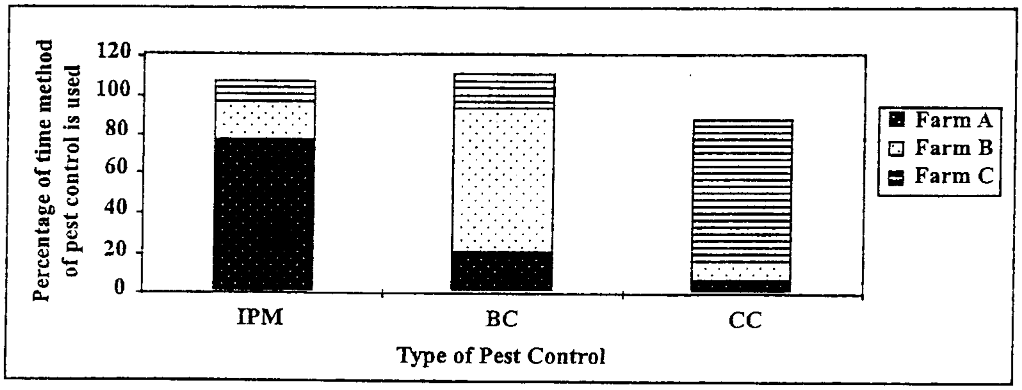

Figure 2. Methods of pest control used by three farms

Based on Figure 2, make FOUR deductions on the use of different methods of pest control by the three farms. **[4 marks]**

(c) Explain, citing TWO reasons, why a decline in agricultural production may be detrimental to the economy of some Caribbean countries. **[6 marks]**

(d) Discuss why EACH of the following is considered an environmentally sustainable agricultural practice:

(i) Agro-forestry **[3 marks]**

(ii) Conservation tillage **[3 marks]**

Total 20 marks

MODULE 2

Answer BOTH questions.

3. (a) Figure 3 shows energy output data from a developed country. Study Figure 3 and use the data it provides to answer the questions that follow.

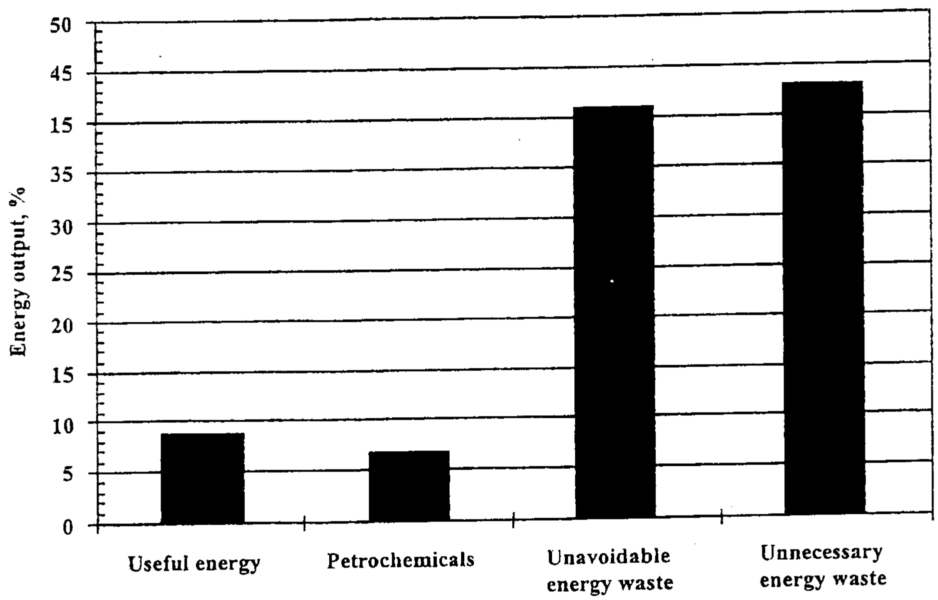

Figure 3. Energy outputs from a developed country

(i) Define the term 'energy efficiency'. **[2 marks]**

(ii) What percentage of this country's energy output is in some beneficial form? Show your working. **[3 marks]**

(iii) Assuming a total energy output of 1 x 10^{15} BTU, how much energy is lost as **unnecessary** energy waste? **[2 marks]**

(iv) The country described in Figure 3 above has a significant amount of energy lost unnecessarily. Suggest THREE ways in which this country could improve its energy efficiency. **[6 marks]**

(b) (i) Define the term 'kinetic energy'. **[1 mark]**

(ii) With the use of a suitably labelled diagram, explain how wind energy is harnessed to produce electricity. **[6 marks]**

Total 20 marks

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4. (a) A small Caribbean island state is considering diversifying its energy sources and using less fossil fuels.

Suggest TWO reasons why the island is considering diversification. Provide ONE economic and ONE environmental reason in your answer. **[4 marks]**

(b) Figure 4 is a diagram of the physical geography of a small Caribbean island. Use the diagram to answer the questions that follow.

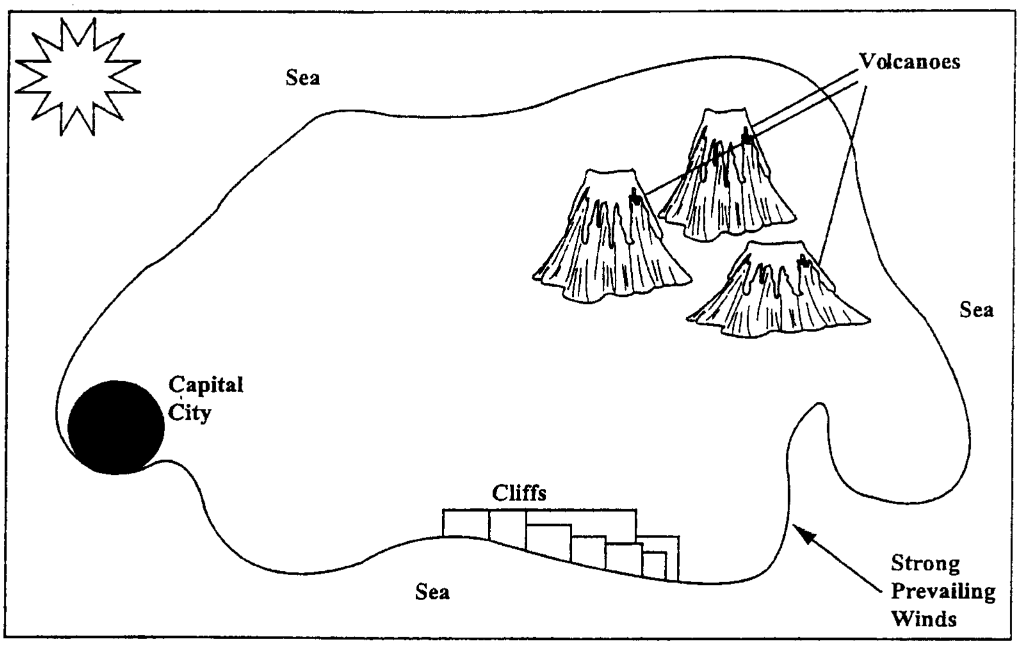

Figure 4. Diagram of the physical geography of a small Caribbean island

(i) Define the term 'renewable energy'. **[2 marks]**

(ii) Based ONLY on the information provided in Figure 4, identify THREE types of renewable energy sources that would be suitable for use on the island. Justify your selection of EACH renewable energy source. **[6 marks]**

(c) Figure 5 depicts the distribution of the commercial energy use of the world (excluding the United States). Study Figure 5 and answer the questions that follow.

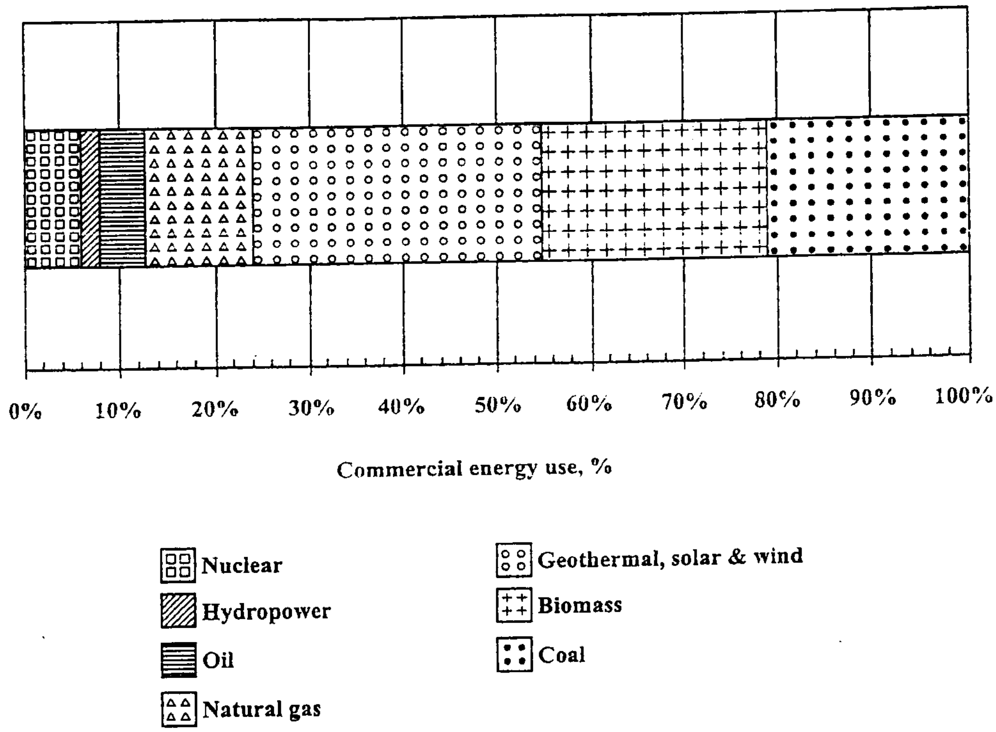

Figure 5. Distribution of commercial energy use by source for the world (excluding the United States)

(i) What percentage of the commercial energy used is derived from renewable energy sources? **[5 marks]**

(ii) In the United States, 7% of the commercial energy used comes from renewable sources. Compare this value with your answer to Part (c) (i), and suggest ONE reason for the difference. **[3 marks]**

Total 20 marks

MODULE 3

Answer BOTH questions.

5. (a) Figure 6 illustrates the results of a two-month monitoring exercise conducted to determine the levels of BOD found in the effluent of a sewage treatment plant. During the course of the monitoring period there were a number of periods when the sewage treatment plant was malfunctioning (not functioning properly). Study Figure 6 below, and use the data it provides to answer the questions that follow.

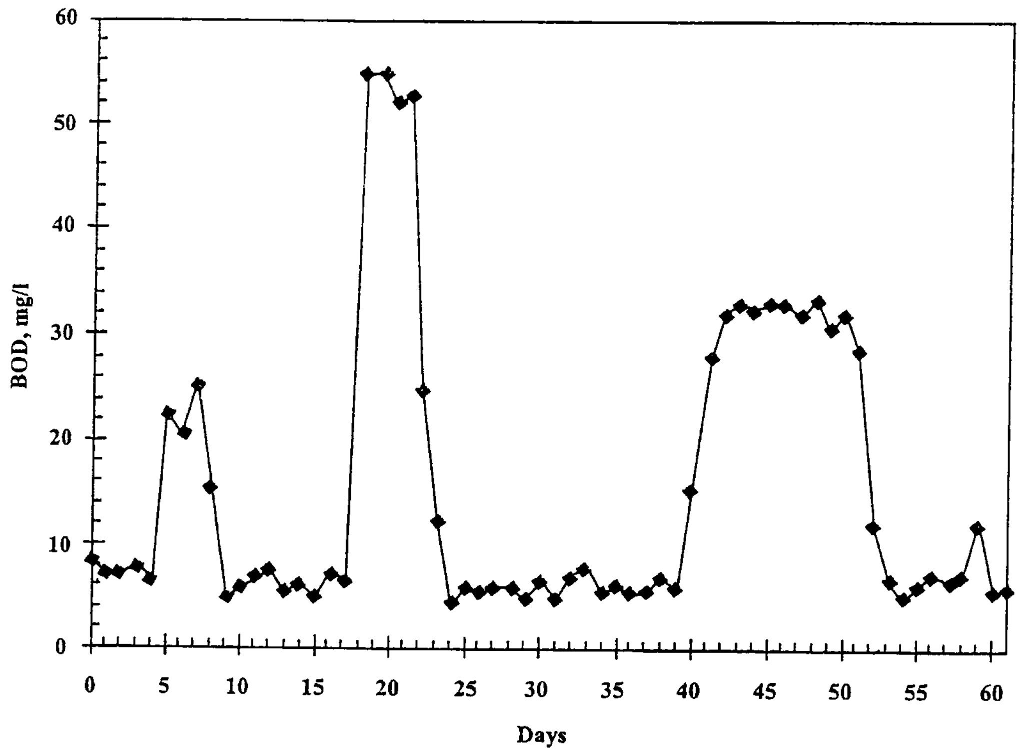

Figure 6. BOD levels in the effluent of a sewage treatment plant over a two-month period

(i) Define the term 'BOD'. **[3 marks]**

(ii) How many times during the monitoring period was the sewage treatment plant malfunctioning? **[1 mark]**

(iii) Explain how you arrived at the answer to (a) (ii). **[2 marks]**

(iv) What were the first and last days of the LONGEST plant malfunction? **[2 marks]**

(v) On average, what was the BOD concentration when the plant was operating normally? **[1 mark]**

(b) Figure 7 illustrates the lower section of a river course. Recently, the oysters collected from the swamp area were found to contain a significant concentration of heavy metals. The river water was analysed at Sample Point A, and the water was found to have very low concentrations of heavy metals. Study the figure and then answer the questions that follow.

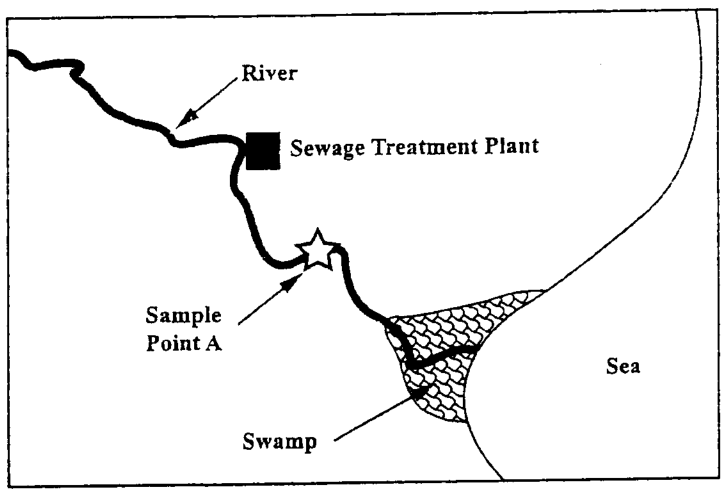

Figure 7. Sketch diagram of the lower section of a river course

(i) Describe the pathway of heavy metal pollution in this section of the river, and use this information to explain why the concentration of heavy metals is much higher in the swamp's oysters than in the river water. **[5 marks]**

(ii) Fully discuss ONE possible environmental impact of the effluent from the sewage treatment plant on the river. **[6 marks]**

Total 20 marks

6. (a) Define the term 'pollutant'. **[3 marks]**

(b) "The institutional framework within a country can be an underlying cause of air pollution." Discuss this statement, including THREE points in your response. **[6 marks]**

(c) (i) Describe the formation of acid rain. **Appropriate chemical equations** are required in your response. **[6 marks]**

(ii) Outline a method to determine if there is an acid rain problem in an entire country. Your outline should provide a description of any experimental techniques you might need and clearly explain how you would use your experimental results to arrive at your conclusion about any acid rain problems. **[5 marks]**

Total 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.